

SCRATCH PROJECT REPORT

Dino Game

1. Introduction

Scratch is a block-based visual programming language developed by the MIT Media Lab. It allows users to create interactive stories, animations, and games using simple drag-and-drop programming blocks. Scratch is especially useful for beginners because it helps develop logical thinking, problem-solving skills, and creativity without requiring prior knowledge of complex programming languages.

This project, titled “**Dino Game**”, is inspired by the popular offline dinosaur running game found in web browsers. The objective of the game is to control a dinosaur character and help it avoid obstacles while scoring points. The game ends when the dinosaur collides with an obstacle, displaying a “Game Over” screen.

The project demonstrates the use of **events, loops, conditionals, variables, broadcasts, sounds, and sprite interactions** in Scratch.

2. Objectives of the Project

The main objectives of this Scratch project are:

- To design an interactive running game using Scratch
 - To understand and apply basic programming concepts such as loops and conditions
 - To use variables for score tracking
 - To implement collision detection between sprites
 - To use broadcast messages for game control
 - To create a user-friendly and visually appealing game interface
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3. Description of the Game

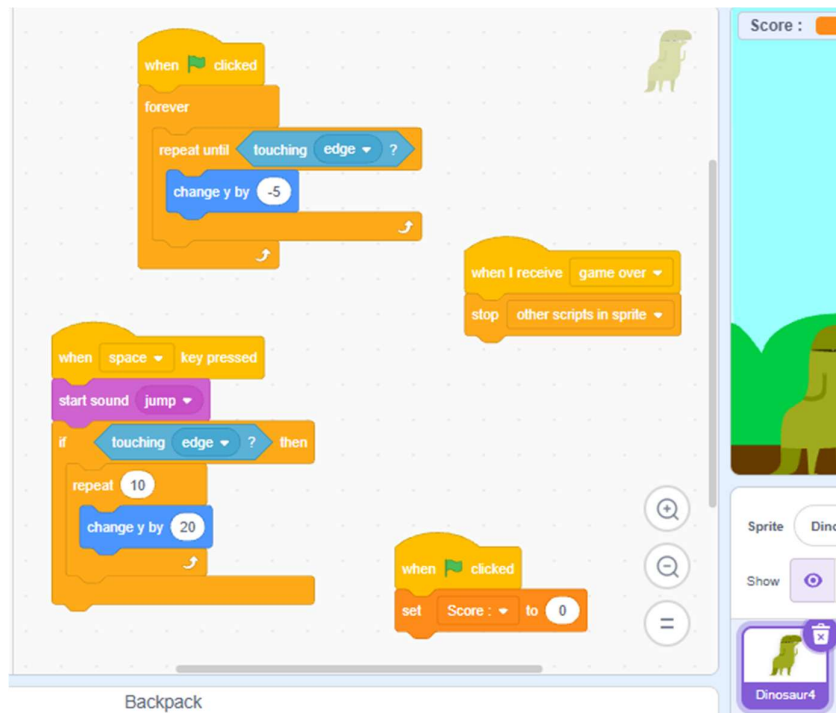
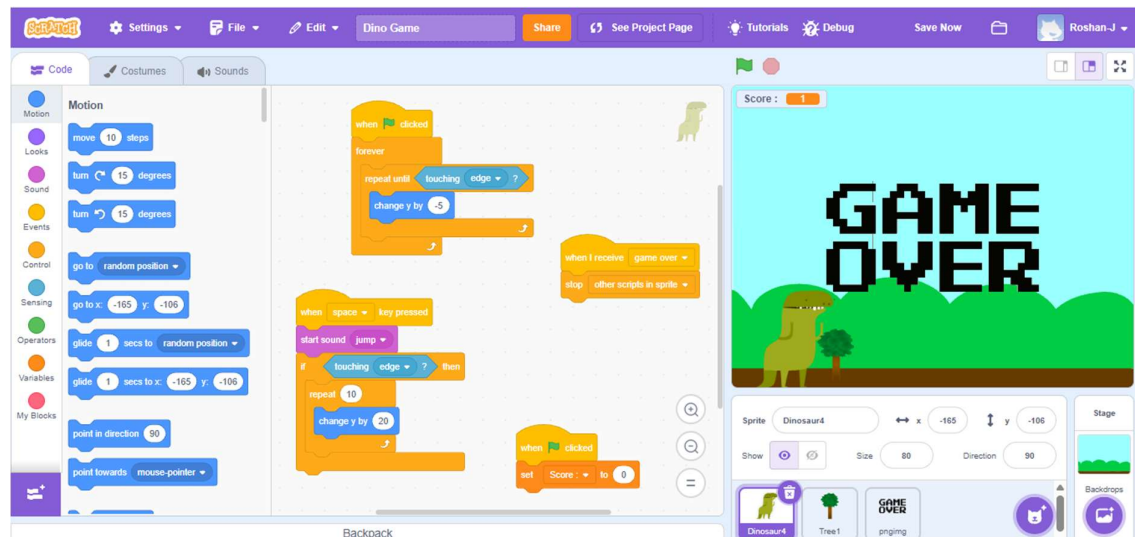
The Dino Game consists of the following main elements:

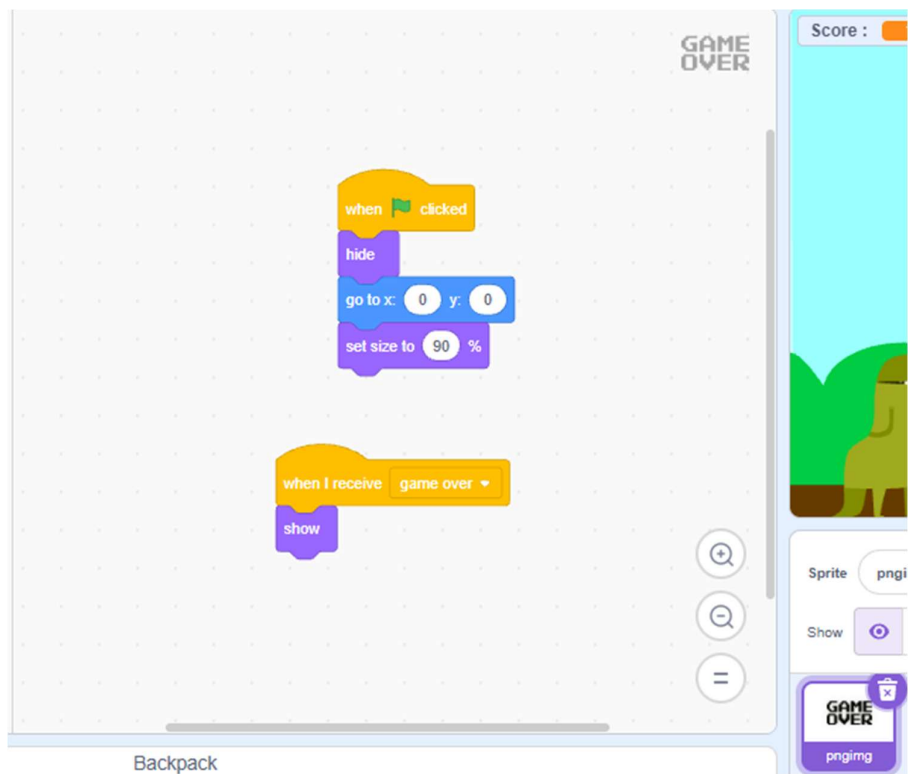
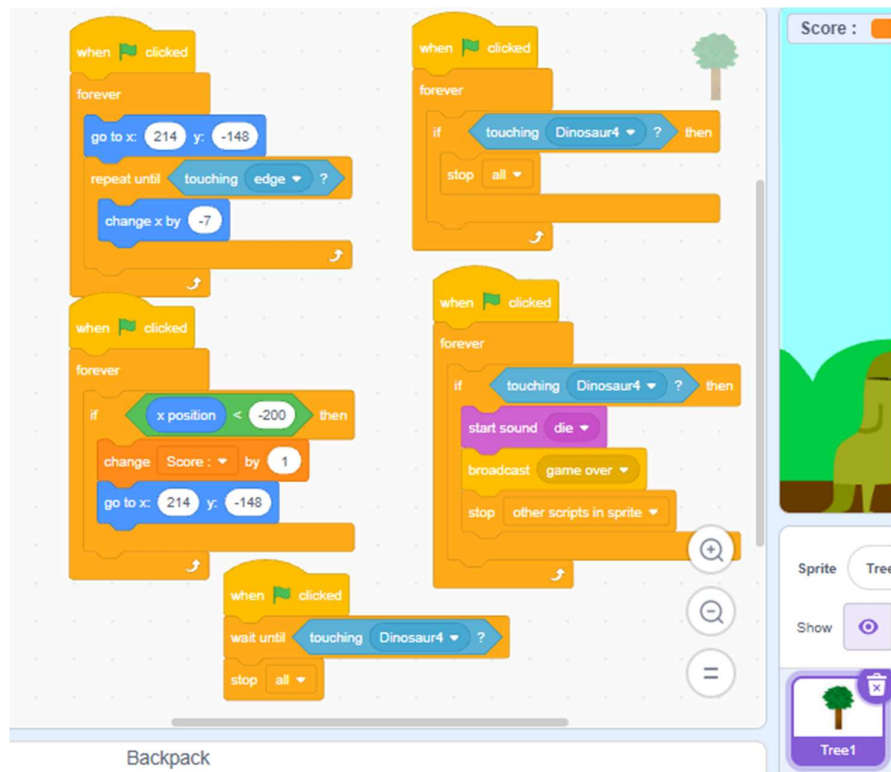
- **Dinosaur Sprite** – the player-controlled character
- **Obstacle (Tree) Sprite** – moves toward the dinosaur
- **Score Variable** – keeps track of the player’s score
- **Game Over Sprite** – appears when the game ends
- **Background (Stage)** – represents the game environment

Game Rules

1. The dinosaur automatically stays on the ground and is affected by gravity.
2. The player presses the **space key** to make the dinosaur jump.
3. Trees move from right to left as obstacles.
4. The score increases each time the dinosaur successfully avoids an obstacle.
5. The game ends when the dinosaur touches a tree.

4. Images





4. Sprites and Their Functions

4.1 Dinosaur Sprite

The dinosaur sprite is the main character of the game and is controlled by the player.

Functions:

- Jump when the space key is pressed
- Fall back down due to gravity
- Detect collision with obstacles
- Stop all actions when the game is over

Important Scripts Used:

- `when green flag clicked` – starts the game
- `forever loop` – applies gravity continuously
- `change y by (-5)` – simulates falling
- `when space key pressed` – makes the dinosaur jump
- `if touching edge` – prevents the dinosaur from going below the ground
- `if touching Tree` – triggers game over

This sprite also plays a **jump sound** and a **die sound**, making the game more engaging.

4.2 Tree (Obstacle) Sprite

The tree sprite acts as the obstacle that the dinosaur must avoid.

Functions:

- Move continuously from right to left
- Reset position after leaving the screen
- Increase the score when successfully passed
- Detect collision with the dinosaur

Important Scripts Used:

- `go to x: 214 y: -148` – places the tree at the starting position
- `repeat until touching edge` – moves the tree across the screen
- `change x by (-7)` – controls obstacle speed
- `if x position < -200` – resets tree and updates score
- `broadcast game over` – sends a message when collision occurs

4.3 Game Over Sprite

The Game Over sprite displays the message “**GAME OVER**” when the dinosaur hits an obstacle.

Functions:

- Remains hidden during gameplay
- Appears only after receiving the game over broadcast

Important Scripts Used:

- `hide` – hides the sprite at the start
 - `when I receive game over` – shows the sprite
 - `show` – displays the Game Over screen
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5. Use of Variables

The project uses a variable called **Score**.

Purpose of the Score Variable:

- Keeps track of how many obstacles the dinosaur has passed
- Increases by 1 each time the tree moves off-screen
- Resets to 0 when the green flag is clicked

Script Used:

- `set Score to 0`
- `change Score by 1`

This makes the game competitive and motivates players to improve their performance.

6. Events and Broadcasts

Broadcast messages are used to control the flow of the game.

Game Over Broadcast

- Sent when the dinosaur touches the tree

- Stops all other scripts
- Displays the Game Over screen

Advantages of Using Broadcasts:

- Keeps the code organized
 - Allows multiple sprites to react to one event
 - Makes the project scalable and easy to modify
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7. Sounds and Visual Effects

The project includes sound effects to enhance the gaming experience:

- **Jump sound** – played when the dinosaur jumps
- **Die sound** – played when the game ends

Visual elements such as pixel-style text and bright backgrounds make the game visually appealing and easy to understand.

8. Testing and Debugging

The game was tested multiple times to ensure:

- The dinosaur jumps correctly
- Obstacles move smoothly
- Score updates accurately
- Game Over triggers at the correct time

Errors such as continuous jumping or incorrect collision detection were fixed by adjusting conditions and loop placements.

9. Limitations of the Project

Some limitations of the current version include:

- Only one type of obstacle
 - No increasing difficulty over time
 - No restart button after Game Over
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10. Future Improvements

The game can be enhanced by adding:

- Multiple obstacle types
 - Increasing speed as score increases
 - Restart button
 - Background music
 - High-score saving feature
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11. Conclusion

The **Dino Game Scratch Project** successfully demonstrates the use of fundamental programming concepts using Scratch. The project helps in understanding game logic, sprite interaction, and event handling in a fun and interactive way. It is an excellent beginner-level game project that can be expanded further with more advanced features.

This project shows how Scratch can be used to create engaging games while learning important programming skills.